

Safe Trajectory Planning Under Uncertainty: A Planner and Solver Comparison on CommonRoad

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Abstract—We compare three trajectory planning formulations for autonomous driving under multi-modal predictions: plan-on-mean, worst-case (minimax), and stochastic chance-constrained. Each planner is paired with two solver backends: Sequential Quadratic Programming (SQP) and the Covariance Matrix Adaptation Evolution Strategy (CMA-ES). These are evaluated on 20 CommonRoad scenarios from NGSIM traffic recordings spanning urban intersections and dense highways. Plan-on-mean is fast but unsafe (34% post-hoc collision rate), as averaging predictions hides dangerous modes. Worst-case CMA-ES is safest (3.8% collision rate) but overly conservative. Stochastic CMA-ES achieves the best balance: 7.5% collision rate, 75% completion, and 15s runtime, while avoiding the extreme lane departures of worst-case planning. SQP fails to converge on over half of all scenarios due to non-smooth objectives and high dimensionality, making CMA-ES the only viable solver for this problem class.

Index Terms—trajectory optimization, stochastic planning, chance constraints, autonomous driving, CMA-ES, sequential quadratic programming, CommonRoad, multi-modal prediction

I. INTRODUCTION

Autonomous vehicles must learn to plan safe and comfortable trajectories in environments where the future behavior of surrounding agents may be uncertain. In situations with dense traffic, this form of uncertainty becomes increasingly challenging as we have more agents in the environment. Nearby vehicles may follow multiple plausible futures (e.g., a car might continue straight, brake, change lanes, or merge into an ego vehicle’s paths). Given these multiple plausible futures, a planner that commits to only one predicted future can thus appear to be safe under its internal model while still producing a trajectory that collides under another plausible mode. Thus, trajectory planning for autonomous driving doesn’t simply require optimizing for smooth vehicle motion, but also reasoning explicitly about multi-modal uncertainty.

A common approach is to collapse multiple predictions into one single deterministic trajectory through the mean predicted obstacle prediction. While this approach can be computationally attractive, it can be unsafe: when prediction modes diverge, the mean trajectory may lie in a region that no actual obstacle future occupies, creating a false sense of clearance. Furthermore, robust or worst-case planners optimize against the most dangerous possible future. These

methods improve safety, but can be overly conservative, which produce uncomfortable or inefficient trajectories that deviate significantly from the lane center. Chance-constrained and stochastic formulations offer a convenient middle ground by planning over a distribution or finite set of possible futures, seeking trajectories that are safe with a high probability while also avoiding the excessive conservatism of pure worst-case planning.

Thus, in this project, we study how planner formulation and solver choice affect the safety, reliability, comfort, and runtime of autonomous driving trajectory optimization under multi-modal uncertainty. We compare three planning formulations. First, a plan-on-mean baseline collapses each obstacle’s four predicted futures into a single average trajectory. Second, a worst-case planner dynamically selects the prediction mode that places each obstacle closest to the actual ego vehicle. Third, a stochastic chance-constrained planner evaluates safety over all prediction modes and accepts trajectories whose empirical collisions risk remains below a threshold. All of the planners we built share the same kinematic bicycle dynamics, smoothness-and-lane keeping objective, and obstacle prediction set, allowing us to isolate the effects for the planning formulation itself. We also compare two solver backends: sequential quadratic programming (SQP) and the Covariance Matrix Adaptation Evolution Strategy (CMA-ES). SQP enforces hard-collision avoidance constraints, but struggles with the non-smooth structure introduced by mode-switching and stochastic collision objectives. CMA-ES is derivative-free and more robust to discontinuities, but handles collision avoidance through penalties rather than explicit constraints. This creates an interesting trade-off between enforcement, convergence reliability, and runtime.

We then evaluate all six planner-solver combinations on 20 CommonRoad scenarios from NGSIM, planning urban intersections and dense highway scenes with 2-44 dynamic obstacle and 1-10.4s horizons. We then report the collision rate, completion rate, trajectory cost, minimum obstacle distance, and runtime.

As we later elaborate upon, our results show that plan-on-mean is efficient, but unsafe and that worst-case CMA-ES is safest, but more conservative, and SQP fails to scale reli-

ably across non-smooth long-horizon scenarios. Thus, overall, stochastic CMA-ES offers the best balance, achieving near-worst case safety with lower cost and runtime.

Our contributions in this work entail a unified comparison of deterministic worst-case, and stochastic planners under shared multi-modal predictions, an empirical comparison of SQP and CMA-ES for non-smooth driving optimization, and a 20-scenario CommonRoad benchmark across urban and highway settings (with an analysis of safety, comfort, reliability, and runtime trade-offs in multi-modal trajectory planning).

II. RELATED WORK

Previous work on trajectory planning under uncertainty ranges from deterministic, robust, and probabilistic formulations. We review these to place our empirical comparison of three planner families across a shared benchmark.

Deterministic planning: Most deployed planners treat trajectory generation as a nonlinear program solved with gradient-based methods. Ziegler et al. [1] used SQP for the Bertha Benz autonomous route; Werling et al. [2] introduced Frénet-frame sampling, later modernized by FRENETIX [3]. These approaches assume a single predicted trajectory per obstacle and rely on safety margins rather than explicit uncertainty reasoning [4].

Robust planning: Minimax formulations guarantee safety by optimizing against worst-case obstacle behavior, but often produce overly conservative or paralyzed behavior in dense traffic [5].

Chance-constrained planning: Blackmore et al. [6] introduced chance constraints for path planning under Gaussian uncertainty. Extensions to multi-modal driving include stochastic MPC with Gaussian mixtures [7], recursively feasible chance-constrained MPC [8], chance-constrained sequential convex optimization [9], and contingency tree planning [10]. Unlike these works, our stochastic planner uses a simple sample-average formulation over $K=4$ discrete prediction modes without distributional assumptions.

Gradient-free solvers: SQP struggles with non-smooth objectives from discrete mode selection. CMA-ES [11] is a derivative-free optimizer suited to such landscapes. We use it as an alternative backend for all three planners, isolating solver choice from planner formulation.

Benchmark: We evaluate on 20 CommonRoad [12], [13] scenarios from NGSIM [14] recordings (Lankershim Boulevard and US-101), using the kinematic bicycle model validated by Kong et al. [15]. To our knowledge, no prior work directly compares deterministic, robust, and chance-constrained planners on the same scenarios with shared prediction sets and evaluation metrics.

III. PROBLEM FORMULATION

A. Vehicle Dynamics

We model the ego vehicle with a kinematic bicycle model. The state and control at timestep t are

$$s_t = [x_t \ y_t \ \psi_t \ v_t]^\top, \quad u_t = [a_t \ \delta_t]^\top, \quad (1)$$

where (x_t, y_t) denotes position, ψ_t is heading, v_t is speed, a_t is longitudinal acceleration, and δ_t is steering angle. The continuous-time dynamics are

$$\dot{s} = f(s, u) = \begin{bmatrix} v \cos \psi \\ v \sin \psi \\ \frac{v}{L} \tan \delta \\ a \end{bmatrix}, \quad (2)$$

where L is the vehicle wheelbase.

We discretize the dynamics using fourth-order Runge-Kutta (RK4), assuming the control input is constant over each timestep Δt . Given s_t and u_t , we compute

$$k_1 = f(s_t, u_t), \quad (3)$$

$$k_2 = f\left(s_t + \frac{\Delta t}{2} k_1, u_t\right), \quad (4)$$

$$k_3 = f\left(s_t + \frac{\Delta t}{2} k_2, u_t\right), \quad (5)$$

$$k_4 = f(s_t + \Delta t k_3, u_t), \quad (6)$$

and update the state as

$$s_{t+1} = s_t + \frac{\Delta t}{6} (k_1 + 2k_2 + 2k_3 + k_4). \quad (7)$$

We use this RK4 rollout to propagate candidate ego trajectories during optimization.

B. Obstacle Prediction Model

For each dynamic obstacle in the scenario, we generate $K=4$ predicted future trajectories representing distinct behavioral modes:

- 1) **Constant velocity:** The obstacle maintains its current speed and heading with zero acceleration.
- 2) **Braking:** The obstacle decelerates at -3.0 m/s^2 until stopped, with no lateral motion.
- 3) **Lane change left:** The obstacle holds its longitudinal speed while drifting laterally at 1.5 m/s over the first 40% of the horizon, following a smooth sinusoidal ramp.
- 4) **Lane change right:** Same as lane change left but in the opposite lateral direction (-1.5 m/s).

Each mode uses a point-mass kinematic model to propagate the obstacle's position over the planning horizon T . The result is a prediction tensor of shape $(K, T, 2)$ per obstacle, giving K possible (x, y) position sequences. All four modes are treated as equally likely. The plan-on-mean planner averages across K ; the worst-case planner selects the closest mode at each timestep; the stochastic planner evaluates collision risk across all K modes simultaneously.

C. Objective and Constraints

All three planners share a common driving cost that penalizes aggressive and jerky maneuvers while encouraging staying in-lane. Given a control sequence $\mathbf{u} = \{(a_t, \delta_t)\}_{t=1}^T$, the objective is

$$J(\mathbf{u}) = \sum_{t=1}^T (a_t^2 + \kappa j_t^2 + \rho d_t^2), \quad (8)$$

where a_t is the longitudinal acceleration, $j_t = (a_t - a_{t-1})/\Delta t$ is the jerk (finite-difference approximation of the acceleration rate of change), d_t is the Euclidean distance from the ego position to the nearest lanelet center line, and $\kappa = \rho = 1$ are weighting parameters. The first two terms promote smooth, energy-efficient driving, while the lane-deviation term discourages off-lane maneuvers.

We formulate collision avoidance differently depending on the solver. For SQP, we impose a hard inequality constraint

$$\min_{i,t} \|\mathbf{p}_t^{\text{ego}} - \mathbf{p}_{i,t}^{\text{obs}}\|_2 \geq r_{\text{safe}}, \quad (9)$$

where $r_{\text{safe}} = 3.0\text{m}$ and the minimum is taken over all obstacles i and timesteps t . For CMA-ES, where constraints cannot be enforced natively, we add a quadratic penalty

$$P(\mathbf{u}) = w_{\text{col}} \sum_{i,t} \max(0, r_{\text{safe}} - \|\mathbf{p}_t^{\text{ego}} - \mathbf{p}_{i,t}^{\text{obs}}\|_2)^2 \quad (10)$$

with weight $w_{\text{col}} = 1000$, giving the unconstrained objective $J(\mathbf{u}) + P(\mathbf{u})$. The quadratic form provides smooth gradients near the constraint boundary while also sharply penalizing large violations.

IV. APPROACH

A. Baseline 1: Plan-on-Mean

Our first baseline plan-on-mean planner reduces the multi-modal prediction problem to a deterministic one by collapsing the $K=4$ predicted futures for each obstacle into a single mean trajectory:

$$\bar{\mathbf{p}}_{i,t} = \frac{1}{K} \sum_{k=1}^K \mathbf{p}_{i,t}^{(k)}. \quad (11)$$

The ego control sequence is then optimized against these averaged obstacle positions using either SQP with the hard collision constraint (9) or CMA-ES with the penalty (10).

This approach is computationally efficient because it reduces K obstacle trajectories to one per obstacle, but it is fundamentally blind to distributional tails. By averaging over prediction modes, the mean trajectory could fall in a region where no individual mode actually places the obstacle, creating a false sense of clearance. In scenarios with divergent modes (e.g., one mode predicts a lane change directly into ego's path while another predicts constant velocity), the plan-on-mean planner optimizes against a phantom "middle" position and can produce trajectories that collide once evaluated under the true modal predictions.

B. Baseline 2: Worst-Case

Our second baseline worst-case planner uses a minimax strategy: at each timestep t , it selects the prediction mode that places the obstacle closest to the current ego position, forming a dynamic adversarial prediction:

$$k_t^*(i) = \arg \min_{k \in \{1, \dots, K\}} \|\mathbf{p}_t^{\text{ego}} - \mathbf{p}_{i,t}^{(k)}\|_2. \quad (12)$$

The collision constraint or penalty is then evaluated against the worst-case positions $\mathbf{p}_{i,t}^{(k_t^*)}$ instead of the mean. Because

the active mode k_t^* depends on the ego trajectory, which itself depends on the optimizer's current control guess, this selection is recomputed at every optimizer iteration.

This formulation is the most conservative of our three planners and exposes a key challenge for gradient-based solvers like SQP. The mode-selection function (12) is piecewise constant: as the optimizer adjusts the ego trajectory, the identity of the closest mode can switch discontinuously. This creates a non-smooth constraint Jacobian that SLSQP cannot handle reliably — the gradient computed in one iteration may contradict the gradient from the previous iteration when a mode switch happens, causing the solver to diverge. CMA-ES, being gradient-free, is immune to this discontinuity and converges reliably even on scenarios where SQP fails entirely. As shown in Table II, SQP converges on only 8–9 of 20 scenarios across all three planners, with failures concentrated in the long-horizon US-101 highway scenes.

The worst-case planner tends to produce overly conservative trajectories that deviate significantly from lane centers, leading to high lane-deviation costs. This over-conservative approach combined with SQP's sensitivity on the non-smooth objective motivates the need for a stochastic planner that balances safety and driving comfort.

C. Stochastic Chance-Constrained Planner

We implemented a stochastic planner which explicitly reasons over all predicted obstacle futures instead of collapsing them into a single mean trajectory or selecting only the most adversarial mode. For each candidate ego trajectory, the planner learns to evaluate whether the ego vehicle is coming within the safety radius of any obstacle under each of the $K=4$ prediction modes. The planner then estimates the collision risk as the fraction of modes that lead to a collision.

The planner enforces a chance constraint by requiring the empirical collision fraction to remain below a tolerance ($\epsilon = 0.1$). We implement this planner with both SQP and CMA-ES. For SQP, we use a smooth approximation of the collision indicator so that the optimizer can evaluate a differentiable chance constraint. For CMA-ES, we add two penalty terms to the objective (one for average collision severity across all prediction modes, and another one for violating the chance constraint). In both of these cases, the planner keeps the same driving objective that is used by the other methods, which penalizes acceleration, jerk, and lane deviation.

For a candidate control sequence u , we evaluate the resulting ego trajectory against all K obstacle prediction modes. For each mode k , we define the closest approach distance as

$$d_k(u) = \min_{i,t} \|\mathbf{p}_t^{\text{ego}}(u) - \mathbf{p}_{i,t}^{(k)}\|_2, \quad (13)$$

where $\mathbf{p}_{i,t}^{(k)}$ is the predicted position of obstacle i at timestep t under mode k . The empirical collision probability is then

$$\hat{p}_{\text{coll}}(u) = \frac{1}{K} \sum_{k=1}^K \mathbf{1}[d_k(u) < r_{\text{safe}}]. \quad (14)$$

The stochastic planner enforces the chance constraint

$$\hat{p}_{\text{coll}}(u) \leq \epsilon, \quad (15)$$

where $r_{\text{safe}} = 3.0$ m and $\epsilon = 0.1$.

D. Solvers

1) *SQP (SLSQP)*: We use SciPy’s SLSQP implementation [16], a gradient-based sequential quadratic programming method that supports bound constraints and nonlinear inequality constraints. The decision variable is the flattened control sequence $[a_1, \delta_1, \dots, a_T, \delta_T] \in \mathbb{R}^{2T}$, with box bounds matching the vehicle limits. Collision avoidance is enforced as a hard inequality constraint (9), and the optimizer is run with `maxiter`=200 and `ftol`= 10^{-6} . The initial guess is zero controls (coast at constant speed). SQP’s main advantage is exact constraint enforcement: when it converges, the resulting trajectory is guaranteed collision-free under the planner’s model. However, SQP requires smooth objective and constraint gradients. This assumption breaks down for the worst-case planner, where mode-switching (12) creates a discontinuous constraint Jacobian, and for long-horizon US-101 scenarios, where the $2T \geq 120$ dimensional search space exceeds the 200-iteration budget. We noticed that SQP is particularly attractive for trajectory optimization because it can directly enforce hard constraints, such as collision avoidance and actuator limits, while optimizing a smooth driving objective. At each iteration, the solver uses local gradient information to estimate how changes in the control sequence affect the future trajectory and constraint satisfaction (this allows SQP to efficiently find low-cost solutions when the optimization problem is smooth and well-conditioned). Note that, SQP relies heavily on accurate gradient information. In our problem formulation, mode switching in the worst-case planner and the non-smooth collision-risk objective in the stochastic planner violate this assumption (which we observed lead to a few convergence failures).

2) *CMA-ES*: The Covariance Matrix Adaptation Evolution Strategy (CMA-ES) [11] is a population-based, gradient-free optimizer that maintains and adapts a multivariate Gaussian search distribution over the decision space. At each generation, CMA-ES samples a population of candidate solutions, evaluates their fitness, and updates the distribution’s mean and covariance matrix to favor promising regions.

We use the `pycma` implementation with an initial step size $\sigma_0 = 0.5$, a fixed evaluation budget of $N_{\text{eval}} = 5000$, and box bounds matching the vehicle’s physical control limits ($a \in [-6, 3]$ m/s², $\delta \in [-0.6, 0.6]$ rad). Because CMA-ES does not support explicit constraints, collision avoidance enters the objective as the quadratic penalty (10) with $w_{\text{col}} = 1000$.

CMA-ES offers two key advantages over SQP for our problem. First, it is robust to non-smooth objectives: the discontinuous mode-switching in the worst-case planner (12) and the non-differentiable collision fraction in the stochastic planner do not affect convergence. Second, its runtime is governed by the fixed evaluation budget instead of the dimensionality of the decision space, making it indifferent to horizon

length. This is a critical advantage on US-101 scenarios with $T \geq 60$ ($2T \geq 120$ decision variables) where SQP exhausts its iteration budget and fails to complete.

The trade-off is that CMA-ES enforces collision avoidance only approximately through the penalty term. The large penalty weight drives the optimizer to maintain clearance exceeding r_{safe} , often pushing the ego vehicle off-lane, which inflates the lane-deviation component of the cost. This explains why the CMA-ES costs are not directly comparable to SQP costs and should be interpreted separately within each solver family.

V. EXPERIMENTS

A. Dataset and Scenarios

We evaluate our planning and solver combinations on 20 real-world driving scenarios from the CommonRoad benchmark. The curated dataset is split equally into two distinct environments: 10 urban road scenarios (Group A) and 10 highway scenarios (Group B). A high-level summary of the selected scenarios can be seen in the Table I.

TABLE I
COMMONROAD EVALUATION SCENARIOS

Scenario ID	Obs.	Duration	Tags
Group A: Lankershim Blvd (Urban Intersection)			
USA_Lanker-1_1	24	4.0 s	lane following, oncoming
USA_Lanker-1_2	42	5.5 s	turn left, oncoming
USA_Lanker-1_3	36	4.0 s	lane following, oncoming
USA_Lanker-1_4	34	1.5 s	turn right, oncoming
USA_Lanker-1_5	35	1.0 s	turn right, oncoming
USA_Lanker-1_6	34	1.0 s	turn left, oncoming
USA_Lanker-1_7	22	1.5 s	lane following, oncoming
USA_Lanker-1_8	31	1.5 s	turn left, oncoming
USA_Lanker-1_9	26	2.0 s	turn left, oncoming
USA_Lanker-1_10	32	1.5 s	turn left, oncoming
Group B: US-101 (Multi-lane Highway)			
USA_US101-1_1	2	6.0 s	lane change
USA_US101-2_1	40	10.4 s	lane change
USA_US101-3_1	34	8.0 s	lane change
USA_US101-4_1	22	10.0 s	lane following, slip road
USA_US101-5_1	25	10.0 s	lane following, traffic jam
USA_US101-6_1	29	8.0 s	lane change
USA_US101-8_1	27	7.5 s	lane change
USA_US101-9_1	30	8.0 s	lane change
USA_US101-10_1	44	8.0 s	lane change, slip road
USA_US101-11_1	32	7.2 s	lane change

We specifically curated this split to ensure a rigorous evaluation across a diverse range of driving conditions:

- **Obstacle Density:** The number of dynamic obstacles ranges widely from sparse (2 agents) to highly dense (44 agents).
- **Planning Horizons:** Scenario durations span from 1.0 s to 10.4 s. This duration range balances short decision windows against long-term planning horizons.
- **Road Types:** Group A focuses on urban navigation, featuring tight intersections and high traffic. Group B tests multi-lane highway dynamics without oncoming traffic.
- **Action Diversity:** The selected split covers a wide range of driving actions including left/right turns, lane following, lane changes, and slip road merging.

B. Evaluation Metrics

To comprehensively evaluate the performance, safety, and efficiency of the three planning formulations across both solvers, we use the following metrics for each of the 20 scenarios:

- **Collision Rate:** The fraction of predicted obstacle futures where the ego vehicle violates the safety radius ($r_{\text{safe}} = 3.0$ m).
- **Completion Rate:** A strict binary indicator that equals 1 only if the optimizer successfully converged and the resulting plan is collision-free.
- **Cost:** The final objective value, which serves as a proxy for driving comfort by summing the squared acceleration, jerk, and lane deviation over the horizon. We evaluate two types of cost: the average cost across all evaluated scenarios, and the average cost strictly isolated to converged-only scenarios.
- **Minimum Distance:** The closest absolute distance the ego trajectory ever gets to any obstacle across all timesteps and all prediction modes.
- **Runtime:** The wall-clock optimization time, measured in seconds, from the first function evaluation to optimizer termination.

C. Results

Table II presents the evaluation results for all six planner-solver configurations across the 20 CommonRoad test scenarios. We report the mean and standard deviation for collision rates, completion rates, objective costs, computation times, minimum obstacle distances, and number of converged scenarios. Figure 1 visualizes these metrics across all configurations.

D. Analysis

Safety (Collision Rate & Minimum Distance): Plan-on-Mean is the least safe of the three planners, with the highest collision rates (33.8%–36.2%). By planning against an averaged obstacle position, it ignores the possibility that any individual prediction mode could place the obstacle directly in the ego vehicle’s path. Interestingly, Plan-on-Mean CMA-ES has the largest average clearance (6.04 m Minimum Distance) yet still collides often (33%); the high variance in obstacle distances means the vehicle occasionally gets dangerously close despite staying far away on average. Worst Case CMA-ES is the safest configuration overall (3.8% collision rate, 5.00 m clearance). Worst Case SQP, despite being a conservative planner, still collides 30% of the time because it only converges on 8 of 20 scenarios, and the resulting trajectories collide frequently. The Stochastic CMA-ES configuration achieves nearly the same safety as Worst Case CMA-ES (7.5% collision rate, 4.98 m clearance), while Stochastic SQP achieves 5.0% collision rate but with lower clearance (4.08 m), reflecting its convergence on harder scenarios (Figures 1a,e).

Reliability (Completion Rate): Solver choice matters far more than planner choice for completion rates. SQP completes only 40%–45% of scenarios across all three planners, because it relies on smooth gradients that break down when the active

obstacle mode switches during optimization, and because it struggles with the large number of decision variables in long-horizon scenarios. CMA-ES, which does not use gradients, avoids both problems and completes 75%–95% of scenarios. Plan-on-Mean CMA-ES achieves the highest completion rate (95%) because averaging the obstacle predictions produces the simplest optimization landscape, making it easiest for CMA-ES to find a good solution.

Driving Comfort (Cost): Cost numbers cannot be directly compared between SQP and CMA-ES because the two solvers handle collision avoidance very differently. SQP uses a hard constraint, so once it finds a valid solution, it is free to minimize the driving cost directly, producing low costs (1.45 for Plan-on-Mean, 4.01 for Worst Case) on the scenarios where it converges.

CMA-ES, on the other hand, penalizes collisions through a large cost term ($w_{\text{col}} = 1000$), which pushes the vehicle away from obstacles but also away from the lane center, inflating reported costs. Within CMA-ES, Plan-on-Mean has the lowest cost (1656.29) because its permissive obstacle positions leave more room for comfortable driving, while Worst Case has the highest (3356.06) because it avoids the most aggressively. One notable result is Stochastic SQP, where the converged-only cost (10,521) is higher than the average across all scenarios (4,870). This happens because SQP only converges on the hardest scenarios, where avoiding all obstacle futures simultaneously forces the vehicle into high-cost maneuvers.

Efficiency (Runtime): CMA-ES is substantially faster than SQP, completing each scenario in 10–38 s on average due to its fixed evaluation budget (Figure 1d). SQP takes 68–398 s depending on the planner. Worst Case SQP is the slowest by far (398 ± 405 s), with a standard deviation larger than the mean, meaning some scenarios take extremely long as the solver repeatedly diverges when the active obstacle mode switches between iterations. This makes Worst Case SQP strictly worse than Plan-on-Mean SQP because it takes $6\times$ longer yet completes fewer scenarios (40% vs. 45%). Within CMA-ES, Worst Case is about $4\times$ slower than Plan-on-Mean (38 s vs. 10 s).

Summary & Best Approach The SQP is not a viable solver for this problem. Across all three planners, it completes fewer than half of the scenarios (40%–45%), takes up to 398 s per scenario, and its costs reflect only the easiest subset of cases where it happened to converge. The non-smooth objectives introduced by mode-switching and the high dimensionality of long-horizon scenarios make it fundamentally unsuitable for real-world deployment.

Within the CMA-ES family, **Stochastic CMA-ES** offers the best overall balance. Plan-on-Mean CMA-ES is faster (10 s vs. 15 s), cheaper (cost: 1656 vs. 3063), and completes more scenarios (95% vs. 75%), but its collision rate of 33.8% makes it unsafe for real deployment. Worst Case CMA-ES achieves the lowest collision rate overall (3.8% vs. 7.5%), and completes more scenarios than Stochastic (85% vs. 75%); however, it has a higher cost (3356 vs. 3063) and longer runtime (38 s vs. 15 s). The safety benefit is marginal: both

TABLE II
COMPARATIVE EVALUATION OF PLANNERS ACROSS 20 SCENARIOS

Planner	Solver	Collision (%)	Completion (%)	Cost (All)	Cost (Conv. Only)	Runtime (s)	MinDist (m)	N(conv)/All
Plan on Mean	SQP	36.2 ± 0.30	45.0 ± 0.50	7.98 ± 17.81	1.45 ± 1.63	76.65 ± 74.08	4.89 ± 2.34	9/20
Plan on Mean	CMA-ES	33.8 ± 0.40	95.0 ± 0.22	1656.29 ± 2115.06	1656.29 ± 2115.06	10.21 ± 5.57	6.04 ± 2.50	20/20
Worst Case	SQP	30.0 ± 0.33	40.0 ± 0.49	68.65 ± 188.58	4.01 ± 4.91	398.63 ± 405.25	3.95 ± 2.44	8/20
Worst Case	CMA-ES	3.8 ± 0.09	85.0 ± 0.36	3356.06 ± 3877.06	3356.06 ± 3877.06	38.07 ± 25.29	5.00 ± 2.16	20/20
Stochastic	SQP	5.0 ± 0.10	45.0 ± 0.50	4869.56 ± 20539.26	10521.69 ± 29649.76	68.90 ± 61.86	4.08 ± 2.59	9/20
Stochastic	CMA-ES	7.5 ± 0.14	75.0 ± 0.43	3063.44 ± 3784.03	3063.44 ± 3784.03	15.12 ± 6.37	4.98 ± 2.31	20/20

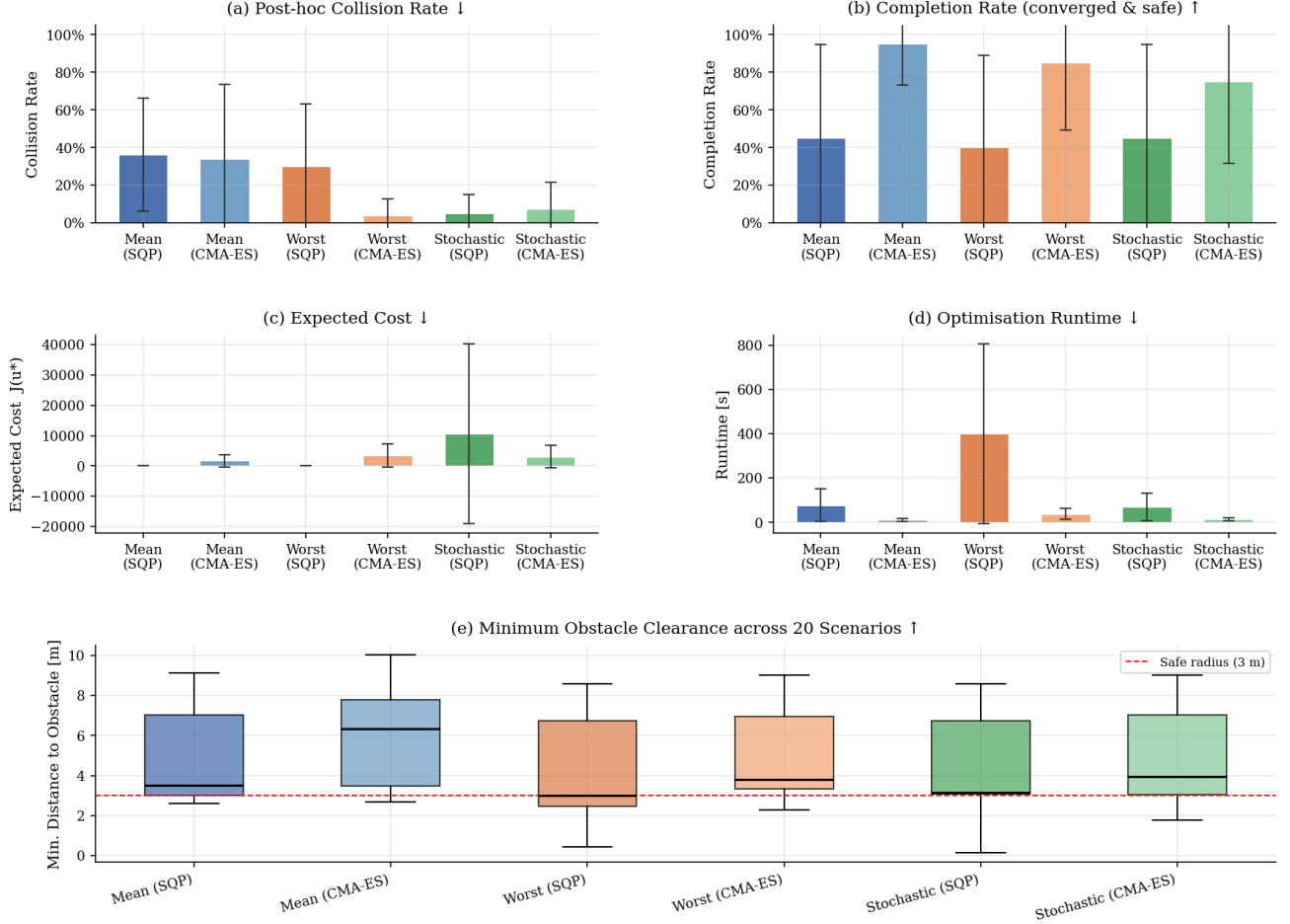


Fig. 1. Planner comparison across 20 scenarios: (a) post-hoc collision rate, (b) completion rate, (c) expected cost, (d) optimisation runtime, and (e) minimum obstacle clearance. The dashed red line in (e) marks the 3 m safety radius. Lower is better for (a), (c), (d); higher is better for (b) and (e).

configurations maintain nearly identical minimum distance clearance (5.00 m vs. 4.98 m), and the 3.7 percentage point difference in collision rate may not justify the comfort and speed penalty. Stochastic CMA-ES produces trajectories that are safe, more comfortable, and faster to compute than Worst Case.

VI. CONCLUSIONS

We presented a comprehensive comparison of three trajectory planning formulations — *plan-on-mean*, *worst-case*, and *stochastic chance-constrained* — paired with two solver backends (*SQP* and *CMA-ES*) across 20 CommonRoad scenarios spanning urban intersections and dense highways. Our

results showed that *SQP* solver is fundamentally unsuitable for this class of problems because of its reliance on smooth gradients which caused it to fail on more than half of all scenarios. *CMA-ES*, by contrast, completes 75%–95% of scenarios reliably across all three planners. Among the *CMA-ES* configurations, Stochastic *CMA-ES* achieves the best overall balance, with a 75% completion rate, 7.5% collision rate, 4.98 m minimum distance clearance, and 15 s runtime. Future work can include exploring hybrid planners that with a combination of stochastic and worst-case objectives and evaluate on longer CommonRoad sequences to assess real-time feasibility.

Beyond comparing planner formulations, our results demonstrate that solver selection is a critical factor in trajectory

optimization under uncertainty where multiple obstacles are present. While the worst-case planning method achieved the lowest collision rates, its additional conservatism increased the cost and runtime. Stochastic planning provided a more practical balance between safety, efficiency, and driving comfort, making it a promising approach for autonomous driving in complex traffic environments. Thus, our findings show that accounting for multi-modal uncertainty is essential for safe autonomous driving. The solver reliability can also have a greater impact on the practical performance than the planner formulation itself (with CMA-ES outperforming SQP on challenging non-smooth scenarios as we observe).

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AUTHOR CONTRIBUTIONS

All authors contributed equally to this work.

Anastasiya Masalava: worked on the CommonRoad dataset curation; kinematic bicycle model with RK4 integration; visualization module (trajectory snapshots, time-lapse grids); comparative evaluation pipeline across all 20 scenarios (collision rate, completion rate, cost, runtime).

Ela N. Sigin: implemented plan-on-mean and worst-case baseline planners with SQP and CMA-ES backends. contributed to final paper sections: related work, approach (plan-on-mean, worst-case, CMA-ES solver), objective and constraints, obstacle prediction model sections. assembled bibliography.

Jaagat Prashar: formulated idea for self-driving planners/solvers with problem formulation, implemented and improved stochastic chance-constrained planner with SQP and CMA-ES backends, contributed to proposal and status update, final paper (Introduction, RK4, SQP, Conclusion, etc.), and appendix.

EXTENSION (4-UNIT)

As our four-unit extension, we added CMA-ES as a gradient-free solver backend for all three planners and conducted a systematic solver comparison against SQP. This required implementing CMA-ES variants of plan-on-mean, worst-case, and stochastic planners with penalty-based collision avoidance, then evaluating all six planner-solver combinations on the full 20-scenario benchmark. The extension revealed that solver choice has a larger impact on reliability than planner formulation: CMA-ES converges on 100% of scenarios (vs. 40–45% for SQP), is 4–40 $\times$ faster, and is robust to the non-smooth objectives that cause SQP to fail on long-horizon and mode-switching problems. The key finding from this extension is that a derivative-free solver fundamentally changes which planning formulations are practical. All authors contributed equally for the extension, ensuring that our work met the requirement for at least 30 additional hours throughout the quarter.

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APPENDIX A ADDITIONAL VISUALIZATIONS

Fig B — Group A (Urban) vs Group B (Highway) Performance Breakdown

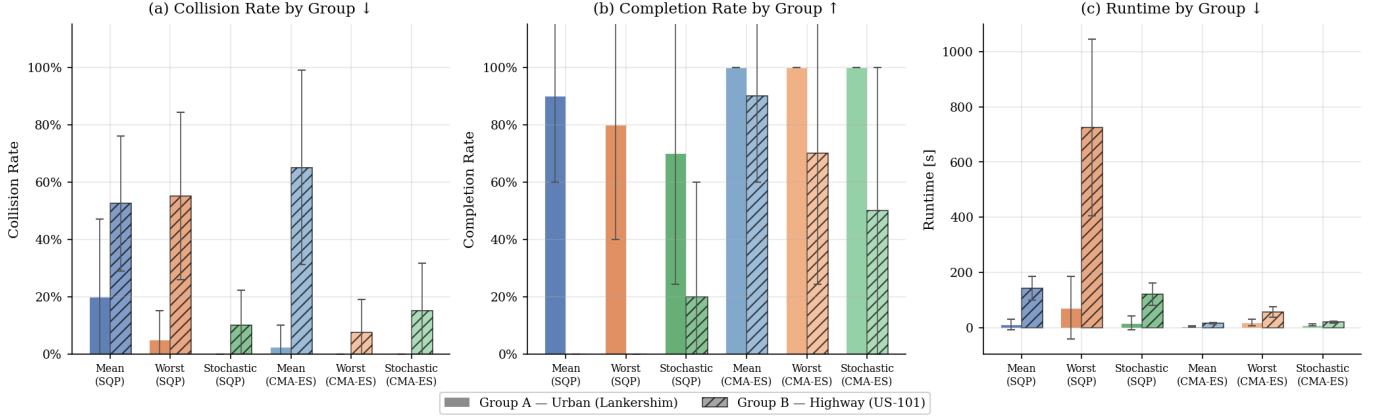


Fig. 2. Urban vs Highway Performance for all solvers and planners

Fig C — Optimisation Runtime vs Planning Horizon

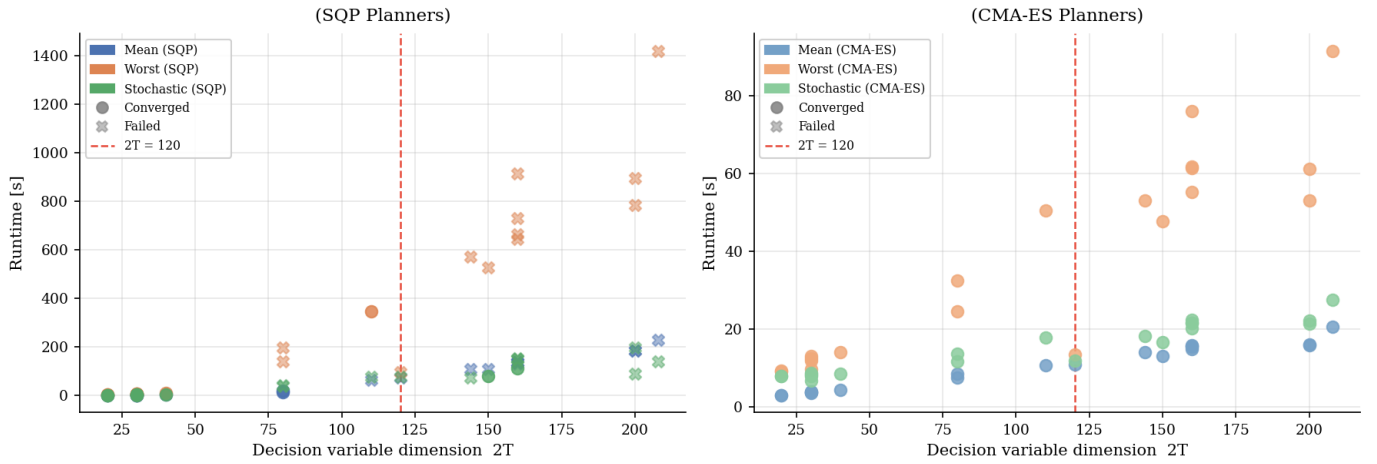


Fig. 3. Runtime comparison for planners SQP and CMA-ES

Optimized ego trajectories — all 6 planner-solver combinations
Scenario: USA Lanker-1 1 T-1 (40 steps, 24 obstacles)

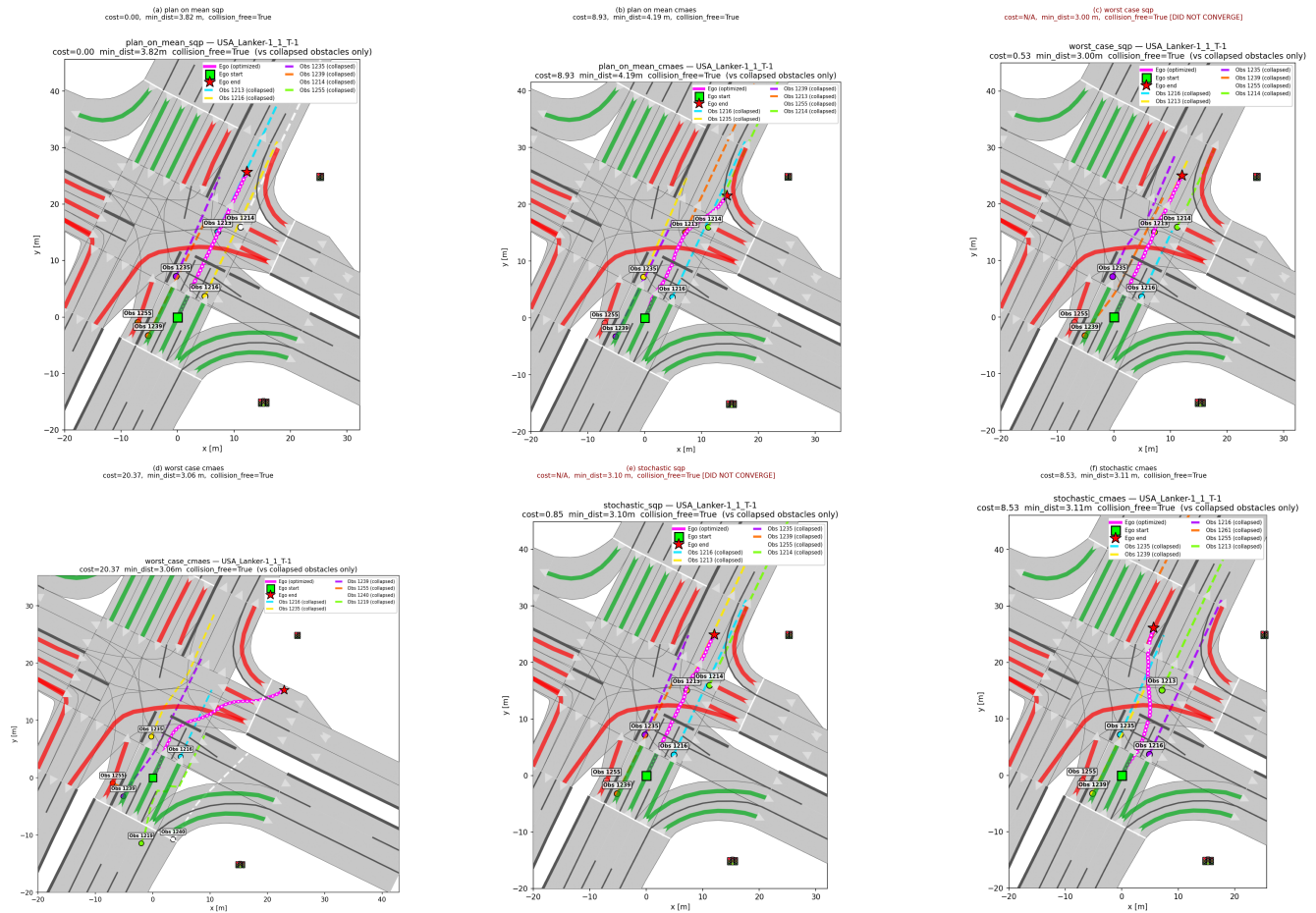


Fig. 4. CommonRoad Optimized Ego Trajectories Example on a Given Scenario